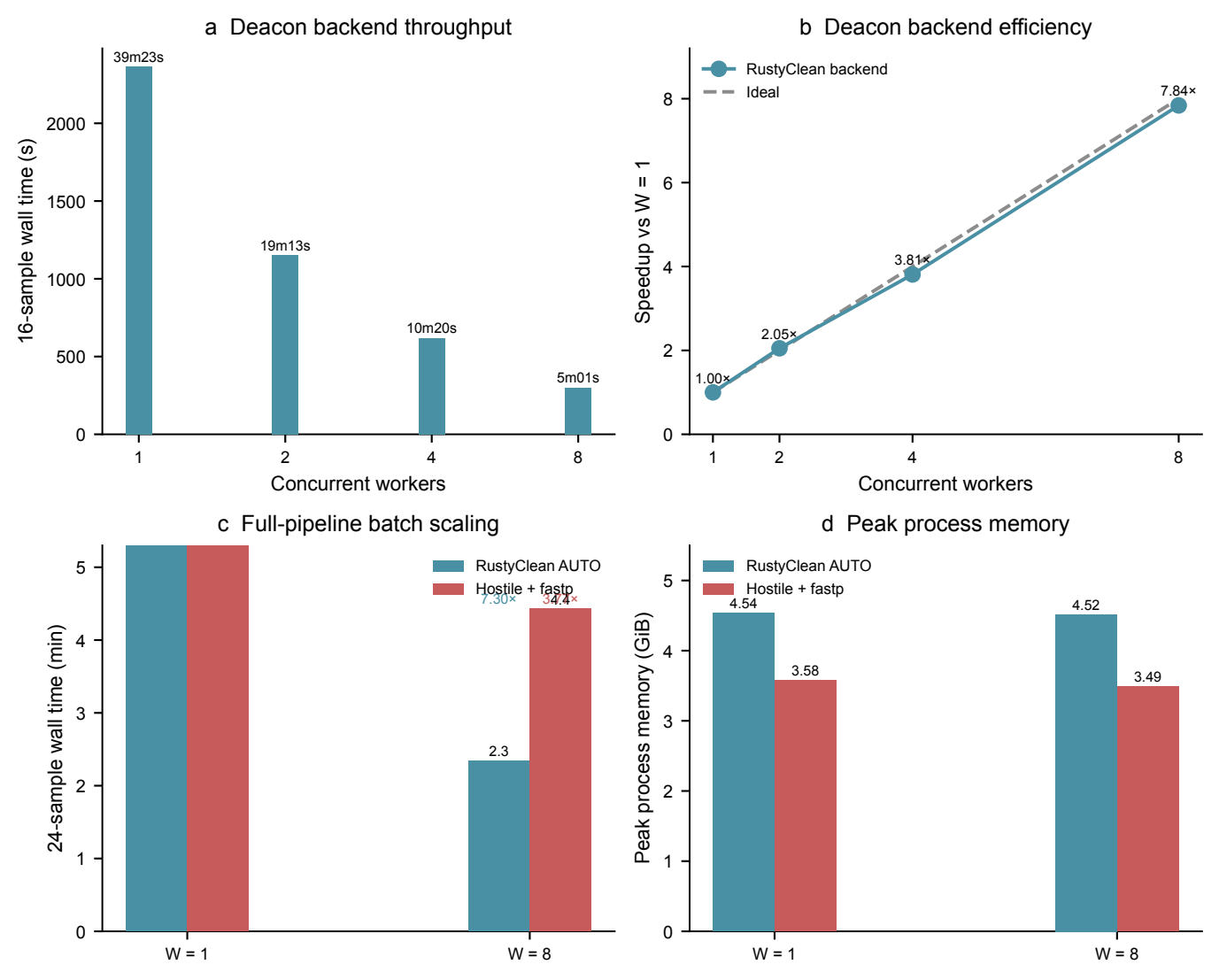
a Deacon backend throughput
16-sample wall time (s)
39m23s
19m13s
10m20s
5m01s
Concurrent workers
b Deacon backend efficiency
RustyClean backend
Ideal
Speedup vs W = 1
1.00×
2.05×
3.81×
7.84×
Concurrent workers
c Full-pipeline batch scaling
RustyClean AUTO
Hostile + fastp
24-sample wall time (min)
7.30×
34.4×
2.3
W = 1
W = 8
d Peak process memory
RustyClean AUTO
Hostile + fastp
Peak process memory (GiB)
4.54
3.58
4.52
3.49
W = 1
W = 8